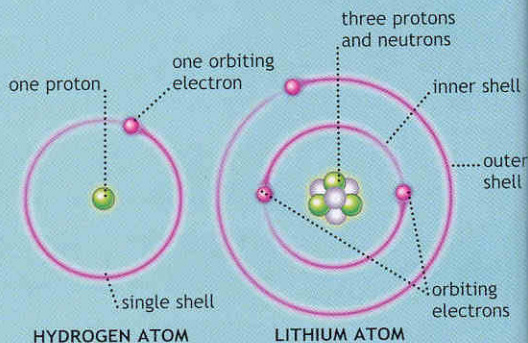


Bohr model

Niels Bohr described the electrons associated with an atom as being in orbits, or shells, around the nucleus. These orbits are in layers, like onion rings, and the quantum

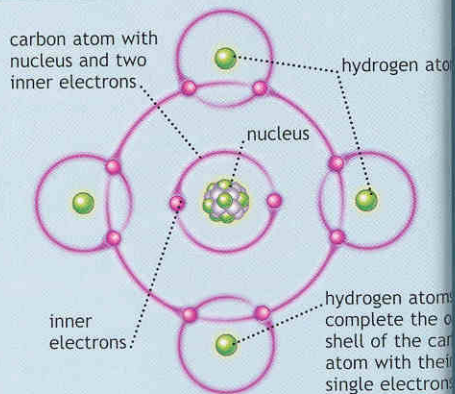
rules only allow a certain number of electrons in each orbit. A full orbit is particularly stable, and atoms will combine with each other to make molecules in such a way that they share electrons to give an illusion of full orbits.

matching protons and electrons
All atoms contain an equal number of protons and electrons. Hydrogen has one proton and one electron, whereas lithium has three of each.



methane

Methane molecules are made of hydrogen and carbon. Hydrogen atoms have one electron, but would "like" to have two, to make a full shell. Atoms of carbon have a full inner shell of two electrons, and a half-full outer shell containing four electrons. So one carbon atom and four hydrogen atoms can get together to make a molecule of methane.



METHANE MOLECULE

Within individual atoms, electrons can jump up to any unfilled shell if they absorb a quantum of energy, and can fall back down, releasing a quantum of energy. This process makes lines in the spectrum of light. The energy levels associated with the shells get closer together the further out they are from the nucleus, like a staircase with steps that get smaller as you go up.

energy exchange
Electrons move from one level to the next by either absorbing or emitting energy.

